

FaceFind — AI-Powered Event Photography · Test Case Repository								
Application Status: 80% Developed Phase: Integration & Acceptance Testing Version: 1.0								
Test ID	Module / Feature	Unit to Test	Assumptions	Test Data	Steps to Execute	Expected Result	Actual Result	Pass/Fail
► AUTHENTICATION TESTS								
TC-AUTH-01	Authentication	Photographer Registration	System is live and accessible	Name: Alex Reynolds Email: alex@moments.com Password: SecurePass@123	1. Navigate to /register 2. Fill all fields 3. Click 'Create Account'	Account created. Confirmation email sent. Redirect to dashboard.	Account created successfully. Confirmation email sent to alex@moments.com. User redirected to photographer dashboard.	Pass
TC-AUTH-02	Authentication	Photographer Login – Valid	Account exists in MongoDB	Email: alex@moments.com Password: SecurePass@123	1. Go to /login 2. Enter valid credentials 3. Click Login	JWT token issued. User redirected to photographer dashboard.	JWT token issued and stored in session. User successfully redirected to photographer dashboard.	Pass
TC-AUTH-03	Authentication	Photographer Login – Invalid Password	Account exists	Email: alex@moments.com Password: wrongpass	1. Go to /login 2. Enter wrong password 3. Click Login	Error: 'Invalid credentials'. No token issued.	Error message 'Invalid credentials' displayed. No JWT token issued. Login page remains active.	Pass
TC-AUTH-04	Authentication	Login – Empty Fields	Login form loaded	Email: (empty) Password: (empty)	1. Submit login form with empty fields	Validation error: 'Username and Password are required'	Validation errors displayed: 'Username and Password are required'. Form not submitted.	Pass
TC-AUTH-05	Authentication	JWT Token Expiry	User is logged in	Expired JWT token	1. Wait for token to expire 2. Attempt to access dashboard	Session expired message. Redirect to /login.	Session expired message displayed. User redirected to /login page automatically.	Pass
► EVENT MANAGEMENT TESTS								
TC-AUTH-06	Authentication	Duplicate Email Registration	Email already in system	Email: alex@moments.com (existing)	1. Attempt to register with existing email	Error: 'Email already in use'.	Error message 'Email already in use' returned. Duplicate account not created in MongoDB.	Pass
TC-EVT-01	Event Management	Create New Event	Photographer logged in	Event Name: Summer Wedding 2025 Date: 2025-06-15 Location: Denver, CO	1. Click 'Create Event' 2. Fill event details 3. Submit	Event created in MongoDB. QR code auto-generated. Event visible in dashboard.	Event 'Summer Wedding 2025' created and stored in MongoDB. Unique QR code auto-generated. Event visible in photographer dashboard.	Pass
TC-EVT-02	Event Management	View All Events	3+ events in DB for photographer	Photographer: alex@moments.com	1. Login 2. Navigate to My Events	All events listed with name, date, photo count, and QR code link.	All events listed with name, date, photo count, and QR code link. Data retrieved correctly from MongoDB.	Pass
TC-EVT-03	Event Management	Event Isolation Per Photographer	Two photographer accounts exist	Photographer A & B each have events	1. Login as Photographer A 2. Check event list	Only Photographer A's events shown. B's events not accessible.	Only Photographer A's events displayed. Photographer B's events not accessible or visible.	Pass
► PHOTO UPLOAD TESTS								
TC-EVT-04	Event Management	Delete Event	Event exists with uploaded photos	Event ID: EVT_001	1. Click delete on EVT_001 2. Confirm deletion	Event, photos, and face embeddings removed from S3 and MongoDB.	Event EVT_001, associated photos, and face embeddings successfully removed from S3 and MongoDB.	Pass
TC-EVT-05	Event Management	QR Code Generation	Event successfully created	Event ID: EVT_002	1. Create event 2. View event details	Unique QR code generated per event, links to guest portal.	Unique QR code generated for EVT_002. QR code correctly links to guest portal for that event.	Pass
TC-PHO-01	Photo Upload	Bulk Photo Upload – Valid Files	AWS S3 configured, event created	50 JPG files, avg 4MB each	1. Select event 2. Upload 50 photos 3. Confirm upload	All photos stored in S3. Face processing triggered via FastAPI. Real-time progress shown via WebSocket.	All 50 photos stored in AWS S3. Face processing triggered via FastAPI. Real-time upload progress displayed via WebSocket.	Pass
TC-PHO-02	Photo Upload	Upload Invalid File Type	Event exists	File: document.pdf	1. Select event 2. Attempt to upload PDF	Error: 'Only JPG, JPEG, PNG files are accepted'.	Error: 'Only JPG, JPEG, PNG files are accepted' displayed. PDF file rejected and not stored.	Pass
TC-PHO-03	Photo Upload	Upload Oversized File	Event exists	File: 50MB RAW image	1. Attempt to upload 50MB file	Error: 'File exceeds maximum size limit'. Upload rejected.	Error: 'File exceeds maximum size limit' displayed. 50MB file upload rejected before transfer.	Pass
► GUEST PORTAL TESTS								
TC-PHO-04	Photo Upload	Real-time Processing Dashboard	WebSocket connection active	30 photos uploaded	1. Upload batch of photos 2. Observe dashboard	Dashboard shows live count of processed photos. Percentage updates in real time.	Dashboard shows live count of processed photos. Percentage updates in real time via WebSocket connection.	Pass
TC-PHO-05	Photo Upload	AI Face Indexing on Upload	ML microservice running (FastAPI)	10 photos with clear faces	1. Upload photos 2. Wait for processing	Face embeddings generated and stored in vector DB. Each face indexed with photo reference.	Face embeddings generated for all 10 photos and stored in vector DB. Each face indexed with corresponding photo reference.	Pass

TC-PHO-06	Photo Upload	Upload Photo with No Faces	Event exists, ML service active	Landscape photo (no people)	1. Upload landscape photo	Photo stored in S3. No face embedding generated. No error thrown.	Landscape photo stored in S3 successfully. No face embedding generated. No error thrown to user.	Pass
TC-GST-01	Guest Portal	QR Code Scan Access	Event is active, QR code valid	Mobile device with camera	1. Scan event QR code 2. Observe redirect	Browser opens guest portal. No app download required.	QR code scan opens guest portal in browser. No app download required. Page loads within 2 seconds.	Pass
TC-GST-02	Guest Portal	Selfie Upload – Valid Face	Guest portal loaded, ML running	Clear front-facing selfie JPG	1. Scan QR 2. Upload selfie 3. Submit	Face matched against event photos. Results returned in < 5 seconds.	Face matched against event photos. All matching results returned in under 5 seconds.	Pass
TC-GST-03	Guest Portal	Selfie Upload – No Face Detected	Guest portal active	Selfie: image of a wall	1. Upload non-face image as selfie	Error: 'No face detected. Please upload a clear selfie'.	Error: 'No face detected. Please upload a clear selfie' displayed. Upload rejected gracefully.	Pass
► FACE RECOGNITION / ML TESTS								
TC-GST-04	Guest Portal	Face Match Returns Results	ML service active, 100 photos indexed	Selfie of person present in 8 event photos	1. Upload selfie 2. Wait for match	All 8 matching photos displayed in gallery. Match accuracy verified.	All 8 matching photos displayed in results gallery. Match accuracy verified above 90% threshold.	Pass
TC-GST-05	Guest Portal	No Match Found	Event photos indexed	Selfie of person not in event	1. Upload selfie of non-attendee	Message: 'No photos found matching your face'. No results shown.	Message: 'No photos found matching your face' displayed. No results shown. No system error.	Pass
TC-GST-06	Guest Portal	Selfie Deleted After Matching	Privacy policy enforced in backend	Valid selfie uploaded	1. Upload selfie 2. Receive results 3. Check S3 for selfie	Selfie not retained in S3 after matching process completes.	Selfie not retained in S3 after matching completes. Privacy policy enforced as expected.	Pass
TC-GST-07	Guest Portal	Expired QR Code / Inactive Event	Event set to expired	QR code from expired event	1. Scan expired QR code	Error: 'This event has ended or is no longer active'.	Error: 'This event has ended or is no longer active' displayed. No access to event portal.	Pass
► CORPORATE EVENT TESTS								
TC-ML-01	Face Recognition	DBSCAN Face Clustering	Python FastAPI ML service running	50 photos, 10 unique individuals	1. Upload 50 photos 2. Trigger clustering	Faces clustered into ~10 groups using DBSCAN. No significant misclassification.	Faces clustered into 10 groups using DBSCAN algorithm. Misclassification rate below acceptable threshold.	Pass
TC-ML-02	Face Recognition	Match Speed < 5 Seconds	100 photos indexed	Selfie of indexed person	1. Upload selfie 2. Start timer 3. Receive results	Results returned in < 5 seconds (SLA requirement).	Match results returned in 3.8 seconds. SLA requirement of < 5 seconds met.	Pass
TC-ML-03	Face Recognition	Match Accuracy – High Similarity	Face embeddings generated	10 clear selfies vs indexed faces	1. Upload 10 test selfies 2. Check match accuracy	≥ 90% of selfies match correct photos.	9 out of 10 selfies matched correct photos (90% accuracy). Meets minimum accuracy requirement.	Pass
TC-ML-04	Face Recognition	Masked/Partially Obscured Face	ML service active	Selfie with face 40% obscured	1. Upload partially obscured selfie 2. Request match	System handles gracefully; returns partial match or 'low confidence' message.	System returned 'low confidence match' for partially obscured face. No crash or unhandled error.	Pass
► SECURITY TESTS								
TC-ML-05	Face Recognition	Multiple Faces in Selfie	ML service active	Group selfie with 3 faces	1. Upload group selfie as guest selfie	Primary face detected. Match attempted for most prominent face.	Most prominent face detected and matched. System returned results without error for group selfie input.	Pass
TC-CORP-01	Corporate Events	Email Whitelist Enforcement	Corporate event with whitelist enabled	Whitelist: ['emp1@corp.com', 'emp2@corp.com'] Test email: external@gmail.com	1. Try to access corporate event as external user	Access denied: 'Your email is not on the approved guest list'.	Access denied: 'Your email is not on the approved guest list' returned for external@gmail.com.	Pass
TC-CORP-02	Corporate Events	Whitelisted Guest Access	Corporate event with whitelist	Email: emp1@corp.com	1. Enter whitelisted email 2. Upload selfie	Access granted. Face match proceeds normally.	Access granted for emp1@corp.com. Face match process proceeded normally.	Pass
TC-CORP-03	Corporate Events	Analytics Dashboard	Event with 50+ guests who accessed portal	Corporate event ID: CORP_001	1. Login as photographer 2. Open CORP_001 analytics	Dashboard shows: total guests, photos accessed, engagement %, export button.	Analytics dashboard loaded for CORP_001. Displays total guests, photos accessed, engagement %, and export button.	Pass
► INTEGRATION TESTS								
TC-CORP-04	Corporate Events	Export Report for Leadership	Analytics data available	Event ID: CORP_001	1. Open analytics 2. Click 'Export Report'	PDF/CSV report generated with engagement metrics downloadable.	PDF/CSV report generated with engagement metrics. File downloadable from analytics dashboard.	Pass

TC-CORP-03	Corporate Events	Session Auto-Expiration	Guest session has 24hr TTL	Guest session token (created 25hrs ago)	1. Attempt to access results with expired session	Session expired. Redirect to QR scan page.	Expired session redirected to QR scan page. Guest prompted to re-scan event QR code.	Pass
TC-SEC-01	Security	SQL/NoSQL Injection in Login	Login endpoint active	Email: {"\$gt": ""} Password: test	1. Enter injection payload in login field 2. Submit	No unauthorized access granted. Input sanitized. Error returned.	Injection payload sanitized. No unauthorized access granted. Error response returned safely.	Pass
TC-SEC-02	Security	CSRF Token Validation	User logged in, form active	Forged request from external domain	1. Submit forged POST request from another origin	Request rejected with 403 Forbidden.	Forged POST request rejected with 403 Forbidden. CSRF token validation enforced.	Pass
► UI/UX TESTS								
TC-SEC-03	Security	Access Other Photographer's Event via URL	Two photographer accounts exist	Photographer A's Event ID accessed by Photographer B	1. Login as Photographer B 2. Manually enter Photographer A's event URL	403 Forbidden returned. No data exposed.	403 Forbidden returned when Photographer B attempts to access Photographer A's event URL. No data exposed.	Pass
TC-SEC-04	Security	Guest Portal HTTPS Enforcement	SSL/TLS configured on server	HTTP URL of guest portal	1. Navigate to http:// version of portal	Automatic redirect to HTTPS. Connection secured.	HTTP URL automatically redirected to HTTPS. SSL/TLS connection established and secured.	Pass
TC-SEC-05	Security	Brute Force Login Protection	Rate limiting enabled	Invalid credentials repeated 10 times	1. Attempt login 10 times with wrong password	Account temporarily locked after 5 failures. Retry after 15 min.	Account temporarily locked after 5 failed login attempts. Retry after 15 minutes message displayed.	Pass
TC-INT-01	Integration	Node.js → FastAPI (ML Service)	Both services running	Photo upload triggering ML pipeline	1. Upload photo via Node.js API 2. Check FastAPI receives request	FastAPI processes photo, returns face embeddings to Node.js within 10s.	FastAPI ML service received photo from Node.js, processed face embeddings, and returned result within 8 seconds.	Pass
TC-INT-02	Integration	S3 Upload & Retrieval	AWS S3 bucket configured	10 test photos	1. Upload via API 2. Retrieve using S3 URL	Photos stored and retrievable with correct permissions.	10 photos stored in S3 successfully. All photos retrievable via S3 URL with correct permissions.	Pass
TC-INT-03	Integration	MongoDB Data Persistence	MongoDB Atlas connected	Create event, upload photo	1. Create event via API 2. Restart server 3. Fetch event	Event data persists after server restart.	Event data persisted after server restart. Fetched event matches originally created event details.	Pass
TC-INT-04	Integration	WebSocket Real-time Updates	WebSocket server running	30 photos uploaded concurrently	1. Connect WebSocket client 2. Trigger upload 3. Monitor messages	Client receives processing updates in real time without polling.	WebSocket client received real-time processing updates for all 30 photos. No polling required.	Pass
TC-INT-05	Integration	Docker Container Communication	Docker Compose running all services	Full stack in containers	1. Start docker-compose 2. Test end-to-end flow	All containers communicate correctly. No connectivity errors.	All Docker containers communicated correctly via Docker Compose network. End-to-end flow completed without errors.	Pass
TC-UI-01	UI/UX	Responsive Design – Mobile	Guest portal deployed	iOS Safari / Android Chrome	1. Open guest portal on mobile 2. Upload selfie	Portal renders correctly on 375px viewport. Selfie upload functions.	Guest portal renders correctly on 375px mobile viewport. Selfie upload functions on both iOS Safari and Android Chrome.	Pass
TC-UI-02	UI/UX	Photo Gallery Display	Guest has matched photos	8 matching photos returned	1. Upload selfie 2. View results gallery	Photos displayed in responsive grid. Download button per photo.	8 matching photos displayed in responsive grid. Download button present per photo.	Pass
TC-UI-03	UI/UX	Loading State During Processing	Match in progress	Selfie submitted, backend processing	1. Submit selfie 2. Observe UI during wait	Spinner/loading indicator shown. No blank screen.	Spinner/loading indicator displayed during processing. No blank screen shown to user.	Pass
TC-UI-04	UI/UX	Admin Dashboard Navigation	Photographer logged in	Dashboard with 3 events	1. Navigate between Events, Analytics, Settings	All sections load without errors. State preserved on navigation.	All sections (Events, Analytics, Settings) load without errors. State preserved on navigation.	Pass